

FOYERS, United Kingdom

WMO: 030570

Lat: 57.267N

Lon: 4.483W

Elev: 21

StdP: 101.07

Time Zone: 0.00 (EUW)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.0	-1.3	-6.3	2.2	-0.5	-4.9	2.5	0.5	12.7	6.1	11.2	7.2	0.5	190	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	22.7	16.3	20.5	15.5	18.8	14.7	17.2	20.8	16.3	19.4	15.3	18.1	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.3	18.7	14.8	10.6	17.6	14.0	10.0	16.9	48.4	21.0	45.6	19.6	43.0	18.1	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.5	8.4	-4.8	25.7	2.3	1.6	-6.5	26.9	-7.8	27.8	-9.2	28.7	-10.9	29.9
			-5.2	19.1	2.2	1.7	-6.7	20.3	-8.0	21.3	-9.2	22.2	-10.8	23.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	4.4	4.6	5.8	7.5	10.0	12.3	14.6	15.0	12.2	9.1	6.6	4.1
	DBStd	4.60	2.67	2.73	2.54	2.52	2.88	2.32	2.04	2.39	2.28	2.68	2.93	3.13
	HDD10.0	921	173	152	131	81	34	6	0	0	5	49	107	183
	HDD18.3	3455	431	384	389	324	257	181	117	108	184	287	353	440
	CDD10.0	513	0	1	1	7	36	76	143	154	71	20	4	1
	CDD18.3	6	0	0	0	0	0	0	2	4	0	0	0	0
	CDH23.3	41	0	0	0	0	3	2	13	22	1	0	0	0
	CDH26.7	2	0	0	0	0	0	0	1	2	0	0	0	0
Wind	WSAvg	3.5	3.8	4.3	3.9	3.6	3.4	3.4	3.5	3.2	3.1	3.1	3.1	3.2
Precipitation	PrecAvg	974	115	87	79	58	63	63	60	74	73	100	94	109
	PrecMax	1359	237	232	205	129	134	140	116	157	212	187	185	252
	PrecMin	636	17	25	22	10	19	21	22	19	21	24	22	17
	PrecStd	156	54	50	43	28	27	30	31	32	38	41	38	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	12.0	13.3	17.2	23.3	22.2	25.2	26.0	21.3	17.4	14.3	12.7
		MCWB	9.6	9.2	8.7	11.6	14.4	14.4	17.2	17.6	16.2	14.2	12.0	10.7
	2%	DB	10.6	10.9	11.5	14.2	18.4	19.6	22.2	22.9	18.6	15.5	12.8	11.4
		MCWB	8.8	9.1	8.7	9.9	12.5	14.0	16.5	17.2	14.5	12.6	11.1	9.7
	5%	DB	9.7	10.1	10.6	12.8	16.0	17.7	19.6	20.6	17.0	14.1	12.0	10.2
		MCWB	7.9	8.3	8.2	9.1	11.6	12.9	15.3	15.9	13.9	11.8	10.2	8.7
	10%	DB	8.5	9.0	9.7	11.6	14.2	16.1	18.1	18.9	15.9	13.0	10.9	8.8
		MCWB	6.9	7.1	7.6	8.6	10.5	12.1	14.7	15.2	13.0	11.0	9.2	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	10.2	10.1	11.9	15.0	15.7	18.3	19.0	17.1	14.8	12.6	11.4
		MCDB	11.2	11.3	11.5	15.5	21.1	19.8	23.0	24.0	20.2	15.9	13.7	12.3
	2%	WB	9.2	9.3	9.3	10.7	13.3	14.4	16.9	17.6	15.5	13.1	11.4	9.8
		MCDB	10.5	10.6	10.9	13.4	17.6	18.2	20.6	21.2	17.8	14.9	12.7	11.2
	5%	WB	8.0	8.4	8.6	9.8	12.0	13.7	15.9	16.6	14.4	12.1	10.4	8.7
		MCDB	9.3	9.9	10.4	12.2	15.3	16.9	18.9	19.9	16.4	13.9	11.8	10.1
	10%	WB	7.0	7.3	7.8	8.9	11.0	12.8	15.1	15.6	13.4	11.2	9.4	7.5
		MCDB	8.5	9.0	9.5	11.1	13.7	15.3	17.7	18.3	15.3	12.8	10.7	8.8
Mean Daily Temperature Range	5% DB	MDBR	3.7	4.1	5.1	5.9	6.6	6.2	6.2	6.5	5.9	4.9	4.0	3.9
		MCDBR	4.7	4.5	6.1	8.4	10.2	9.2	9.4	9.6	7.8	6.3	4.9	5.6
		MCWBR	4.3	3.8	4.0	5.0	5.1	4.8	4.9	4.8	4.5	4.5	4.2	4.8
	5% WB	MCDBR	4.6	4.5	5.5	6.6	9.3	8.0	8.3	8.7	6.9	5.7	4.9	5.3
		MCWBR	4.4	4.1	4.3	4.7	5.1	4.8	4.8	4.9	4.7	4.8	4.5	4.7
Clear-Sky Solar Irradiance	taub	0.324	0.344	0.368	0.388	0.393	0.387	0.388	0.372	0.359	0.344	0.322	0.313	
	taud	2.170	2.243	2.242	2.242	2.284	2.341	2.350	2.401	2.430	2.380	2.248	2.127	
	Ebn at Noon	513	686	774	820	842	853	843	832	786	686	520	407	
	Edh at Noon	55	78	101	118	120	115	112	100	84	68	51	44	
All-Sky Solar Radiation	RadAvg	0.38	0.99	2.00	3.37	4.51	4.38	4.15	3.30	2.30	1.21	0.52	0.28	
	RadStd	0.06	0.13	0.17	0.26	0.37	0.45	0.43	0.24	0.18	0.14	0.05	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page